Jonathan Setiawan

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SUMMARY / PROFESSIONAL OBJECTIVE

Robotics and AI Researcher with interest in practical application of autonomous systems within high-impact sectors. Proven expertise in engineering robust perception-to-action pipelines and software for real-world deployment.

EDUCATION

Ritsumeikan University (立命館大学) — **Osaka, Japan**

College of Information Systems Science and Engineering — *Graduation Date: April 2028*

Awards: Murai SIIX Scholarship (2025), JASSO Monbukagakusho Honors Scholarship (2024)

Certifications: IELTS 8.0 (2022) — *CGPA: 4.35/5.0*

Relevant Coursework: Network systems, Artificial Intelligence, Team Project Based Learning, C Programming, DSA

SKILLS

Python, C, C++, Java, JavaScript/TypeScript, MySQL, ROS2 (Humble), Gazebo, Networking, OpenCV, BERT, PyTorch, TensorFlow, SLAM, Docker, Git, React, CMake, LaTeX, Linux (Ubuntu, Fedora), Go, PHP, Rust

RESEARCH & LEADERSHIP EXPERIENCE

iCOM技研株式会社 — **Hyogo, Japan**

R&D Systems Engineer Intern — *Jul 2025 - Present*

UR5e Pick and Place System Project

- *Architected a palletizing and depalletizing system for UR5e, UR10e, and UR20 robots, managing DHCP protocol and network configurations.*
- *Engineered a high-stability box size classification system using RANSAC-based surface modeling and temporal filtering, achieving ≥0.97 confidence score while mitigating segmentation noise via mask erosion.*
- *Automated industrial hand-eye mapping using Tsai-Lenz (AX=XB) methodology and the RTDE API, reducing spatial error to <2mm and implementing weight-bias heuristics for stable, continuous sorting operations.*

Ritsumeikan University (立命館大学) — **Osaka, Japan**

RiOne (Robocup @Home League) Robotics Club Computer Vision Team Lead — *Jul 2024 - Present*

- *Architected a multi-node ROS 2 Humble perception stack integrating face re-identification and temporal tracking to enable persistent Human-Robot Interaction (HRI) in dynamic environments.*
- *Engineered a closed-loop object recognition pipeline that autonomously logs detection data and retraining metrics, creating a self-improving vision system for robust deployment*
- *Deployed a sensor-fused navigation stack on TurtleBot3 WafflePi for JapanOpen 2025, using Intel RealSense depth data with SLAM for autonomous obstacle avoidance and danger detection.*

PROJECT EXPERIENCE

SciBERT-CTFT: Scientific Abstract Classifier via Contrastive Learning Fine-tuning

Artificial Intelligence Course Project — *Dec 2025 - Jan 2026*

- *Fine-tuned a SciBERT model using the SetFit framework to classify scientific literature with 99.61% accuracy, implementing a novel "Context Injection" data augmentation technique using TF-IDF maps.*
- *Validated model robustness via 5-fold cross-validation, demonstrating consistent performance (>0.99 F1-score) across domains like Bioinformatics and Neuroscience despite limited labeled data.*

EFT-VPR: Event Forecasting Transformer for Visual Place Recognition

Personal Project — *Feb 2026 - Mar 2026*

- *Engineered an event-based localization stack utilizing a 3-layer SNN encoder and an 8-head autoregressive Transformer with InfoNCE contrastive learning, improving Recall@1 by 13.8% in a 65,992-frame dataset.*
- *Designed a production-ready VPR pipeline leveraging FAISS (IndexFlatIP/GPU) for top-K cosine retrieval and dropout-robust forecasting, reducing median localization error by 27.8% (81.9m to 59.1m).*

AESP: Adaptive Event Signal Processing

Personal Project — *Feb 2026 - Mar 2026*

- *Architected a Numba-accelerated spatio-temporal Poisson filtering pipeline for high-entropy event-based camera streams, delivering a +13.82 dB SNR gain with 99.14% signal preservation for stable downstream transformer forecasting.*
- *Engineered a production-stable adaptive controller using hysteresis (Fast-Tighten/Slow-Relax) on EPPS-driven thresholds, reducing event bandwidth by 37.80% (69,511 to 43,237 events) while sustaining a 67.32 ms real-time processing latency.*